Be Bayesian my Friend. Part I: An Introduction to Bayesian Learning

IV Congreso de Usuarios de R

Joaquín Martínez-Minaya, November 5, 2025

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What do the following events have in common?

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Palomares Bomb

- In a routine **Cold War** operation on 17 January 1966 in Palomares (Almería, Spain), an incident happened.
- At 10:22 a.m. that day, **two aircrafts collided over the skies of Almeria**: a KC-135 from Moron Air Base (Seville) was about to refuel a B-52 from Turkey that was returning to North Carolina.
- Only **4 of the 13** crew members of both aircraft survived.
- The second plane returning from Turkey contained 4 atomic bombs. And all four were going to fall squarely on Palomares in the east of Almería.
- Of the **four bombs**, three fell on land and could be located. Numerous witnesses assured the US Army detachment to Palomares that the **stray bomb** had landed in the water. But, **how to find it?**

The casings of two B28 nuclear bombs involved in the Palomares incident, on display at the National Atomic Museum, in Albuquerque

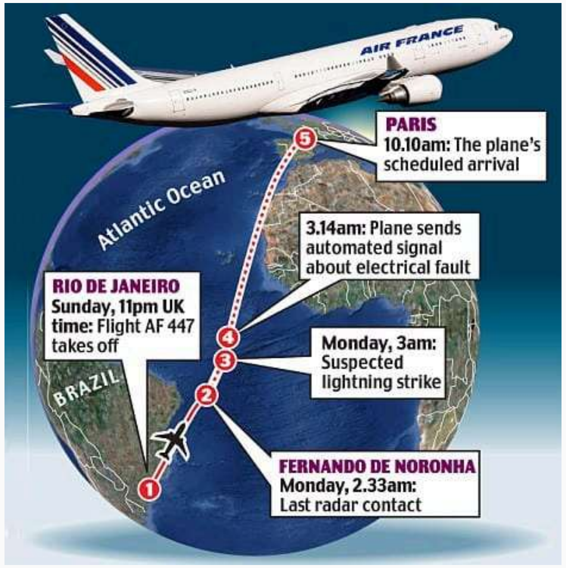
Alan Turing and Enigma code

- During the Second World War II, Alan Turing and his team worked on decrypt the **enigma code**.
- Algorithm developed was called **"bamburismus"**. This was used to **decrypt messages sent by the German Navy**.
- The Enigma machine consisted of a keyboard, a panel where the letters lit up and **several rotors**.
- To encrypt a message, the rotors were placed in a **certain position** and the message was written and the encrypted message was displayed on the panel.
- To decrypt an encrypted message, the process was symmetrical. Simply **set the rotors to the initial configuration** and type in the encrypted message, which would appear decoded on the panel. But, **how bamburismus worked to decrypt messages?**

Enigma machine

Air France Flight 447

- On **1 June 2009**, Air France Flight 447 disappeared while flying from **Rio de Janeiro to Paris**. The aircraft vanished over the **Atlantic Ocean** with **228 people** on board.
- For **two years**, all search attempts failed. The ocean was vast, the information fragmented, and every new hypothesis seemed to contradict the previous one.
- Investigators decided to change their strategy: instead of searching blindly, they **combined every piece of available information** —last radar signals, ocean currents, debris drift patterns, and expert judgement.
- As new traces appeared, these clues **reshaped their belief** about where the aircraft could be found.
- But, **how did they manage to search so efficiently in such uncertainty?**



Flight path and last known signals of Air France Flight 447 (Rio de Janeiro – Paris, June 1, 2009)

How these problems were solved?

Palomares Bomb

- Experts **assigned probabilities to each area of the map** and **updated them** as new information arrived. Guided by **Paco, a fisherman who was near the impact point**, they **redrew the map** and finally located the bomb.

- Years later, **Lawrence D. Stone formalised this idea as the Bayesian Search Model**.

Alan Turing and Enigma code

- As coded messages were received, the belief about the **hypothetical machine configuration was updated**.

- When the **weight of evidence in favour of a particular configuration** of the Enigma machine was sufficiently high, that configuration was considered probable.

Air France Flight 447

- Investigators built a **Bayesian search model** combining radar, ocean, and expert data. As new debris and signals appeared, **probabilities were updated sequentially**.
- The wreck was found within the area of **highest posterior probability**.

Common principle

Different decades, same idea:

Each problem was solved by **combining prior knowledge with new evidence**,
and by **updating beliefs probabilistically**.

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1. A bit of history

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Thomas Bayes (1701-1761)

- He was an **English statistician, philosopher and Presbyterian minister**.

- Specifically, he was the **eldest of 7 children** born to Ann and Joshua Bayes. Joshua was one of the first seven Presbyterian reverends, a branch of the Protestant church that did not support the Anglican church.

- He moved to London, but as a Presbyterian, **Bayes was unable to study theology** in that city, and had to travel to the city of Edinburgh to study theology. It seems that he studied also **Mathematics and Logic**.

- Thomas was interested in studying **the most probable causes** of what was happening in order to prove God's existence and benevolence.

The work that changed everything

- In this respect, one of the people Thomas most admired was **Sir Isaac Newton**. Newton was trying to prove that if **there were such rules, it was because there was a God who organized everything**.

- Bayes devises a procedure by which, **starting from, zero knowledge, one can learn from observations (data, facts) to get to know what causes them**.

- During his lifetime, he did not want to **publish his theorem** because he thought it was irrelevant.

- It only saw the light of day when his friend **Richard Price** retrieved it from a pile of papers. Finally, it appeared in 1763, in the journal Philosophical Transactions under the name: **An Essay toward solving a Problem in the Doctrine of Chances**.

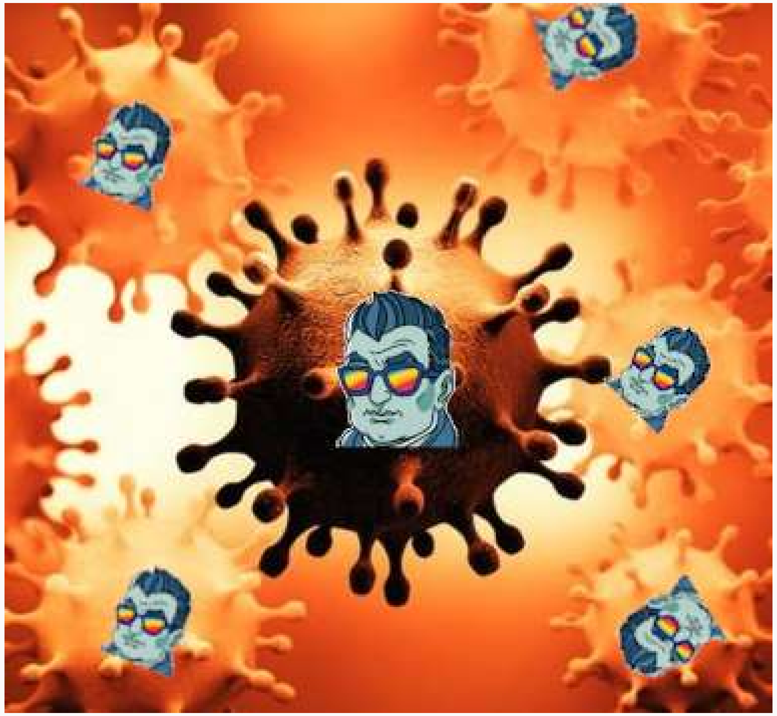
How does it work?

- I know something about the hypothesis -> **Prior distribution**: $P(Hypothesis)$.
- I observe the data -> **Likelihood**: $P(Data \mid Hypothesis)$.
- I update my knowledge about my hypothesis -> **Posterior distribution**: $P(Hypothesis \mid Data)$.

$$P(Hypothesis \mid Data) = \frac{P(Data \mid Hypothesis) \cdot P(Hypothesis)}{P(Data)}$$

Example: Bayesianitis

- A test for detecting **Bayesianitis** (a Statistics condition) has been made. This test has:
    - 95 % sensitivity
    - 98 % specificity
- It is known that **1 of 100 statisticians** has this condition. The test is given massively in different universities in Spain. What is the Probability that Laplace, who has **tested + in this test is indeed Bayesianitis +**?
- Laplace has tested positive in the first test, what would be the probability to be **Bayesianitis + if Laplace tests positive again?**



"Bayesianitis" — a fictional condition representing the contagious nature of Bayesian reasoning.

For an intuitive explanation of *sensitivity* and *specificity*, see El País, *Café y Teoremas* (2021)

Example: Bayesianitis. Posterior

Hypothesis. Prior information

B is 'true condition' of Laplace:

- B^+: Laplace is Bayesianitis +
- B^-: Laplace is Bayesianitis -

"1/100 prevalence" -> $P(B^+) = 0.01$

Data. Observed and then, known

$+_1$ the first test is positive.

95% sensitivity -> $P(+_1 \mid B^+) = 0.95$

98% specificity -> $P(-_1 \mid B^-) = 0.98$

Updated Hypothesis. Posterior information

$$P(B^+ \mid +_1) = \frac{P(+_1 \mid B^+)P(B^+)}{P(+_1)}$$

$$= \frac{P(+_1 \mid B^+)P(B^+)}{P(+_1 \mid B^+)P(B^+) + P(+_1 \mid B^-)P(B^-)} =$$

$$= \frac{0.95 \times 0.01}{0.95 \times 0.01 + 0.02 \times 0.99} = 0.324$$

- **32.4% of those Statisticians getting a `(+)` test are Bayesianitis +.**

Example: Bayesianitis. Testing again. Updating

- What would be the probability to be Bayesianitis+ if Laplace tests positive again?

Hypothesis. Prior information

Prior information now is posterior of the previous process.

$$P(B^+ \mid +_1) = 0.324$$

Data. Observed and then, known

$+_2$ the second test is positive.

95% sensitivity -> $P(+_2 \mid B^+) = 0.95$

98% specificity -> $P(-_2 \mid B^-) = 0.98$

Updated Hypothesis. Posterior information

$$P(B^+ \mid +_2, +_1) = \frac{P(+_2 \mid B^+, +_1)P(B^+ \mid +_1)}{P(+_2 \mid +_1)}$$

$$= \frac{P(+_2 \mid B^+)P(B^+ \mid +_1)}{P(+_2 \mid B^+, +_1)P(B^+ \mid +_1) + P(+_2 \mid B^-, +_1)P(B^- \mid}$$

$$= \frac{0.95 \times 0.32}{0.95 \times 0.32 + 0.02 \times 0.68} = 0.958$$

- **95.8% of those Statisticians getting a second `(+)` test are Bayesianitis +.**

Example: Diabetes Screening

- A test for detecting **Diabetes** (a health condition) has been conducted. This test has:
  - 80 % sensitivity
  - 97 % specificity
- It is known that **14.8% of individuals** have this condition. The test is administered widely in different healthcare facilities in Spain. What is the probability that an individual who has **tested + in this test indeed has diabetes +**?
- The individual has tested positive in the first test, what would be the probability of having **diabetes + if they test positive again?**

Example: Diabetes Screening. Posterior

Hypothesis. Prior information

D is 'true condition' of the individual:

- D^+: Individual has diabetes
- D^-: Individual does not have diabetes

"14.8% prevalence" -> $P(D^+) = 0.148$

Data. Observed and then, known

$+_1$ the first test is positive.

80% sensitivity -> $P(+_1 \mid D^+) = 0.80$

97% specificity -> $P(-_1 \mid D^-) = 0.97$

Updated Hypothesis. Posterior information

$$P(D^+ \mid +_1) = \frac{P(+_1 \mid D^+)P(D^+)}{P(+_1)}$$
$$= \frac{P(+_1 \mid D^+)P(D^+)}{P(+_1 \mid D^+)P(D^+) + P(+_1 \mid D^-)P(D^-)} =$$
$$= \frac{0.80 \times 0.148}{0.80 \times 0.148 + 0.03 \times 0.852} = 0.822$$

- **82.2% of individuals getting a `(+)` test have diabetes.**
- **What would be the probability for the individual to be diabetic if we conduct a second test?**

Example: Transition to Bayesian Distributions

- We now apply the Bayes theorem to statistical models by changing the notation to use a parameter θ, which represents the true condition:
  - $\theta = 1$ for D^+ (disease present)
  - $\theta = 0$ for D^- (disease absent)
- Test results are recoded as data:
  - $y = 1$ indicates a positive test result -> T^+
  - $y = 0$ indicates a negative test result -> T^-
- **The posterior probability of the individual having the disease given a positive test result is**:

$$p(\theta = 1 \mid y = 1) = \frac{p(y = 1 \mid \theta = 1) \cdot p(\theta = 1)}{p(y = 1 \mid \theta = 1) \cdot p(\theta = 1) + p(y = 1 \mid \theta = 0) \cdot p(\theta = 0)}$$

- $p(\theta = 1)$ represents the prior probability of disease (prevalence), while $p(y = 1 \mid \theta = 1)$ is the likelihood function for a positive test.

2. Bayesian approach

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Density and probability functions

Bayes Theorem

$$P(Hypothesis \mid Data) = \frac{P(Data \mid Hypothesis) \cdot P(Hypothesis)}{P(Data)}$$

Bayesian Inference

$$p(\boldsymbol{\theta} \mid \boldsymbol{y}) = \frac{p(\boldsymbol{y} \mid \boldsymbol{\theta}) \cdot p(\boldsymbol{\theta})}{p(\boldsymbol{y})} = \frac{p(\boldsymbol{y} \mid \boldsymbol{\theta}) \cdot p(\boldsymbol{\theta})}{\int p(\boldsymbol{\theta}) p(\boldsymbol{y} \mid \boldsymbol{\theta}) \mathrm{d}\boldsymbol{\theta}} \propto p(\boldsymbol{y} \mid \boldsymbol{\theta}) \cdot p(\boldsymbol{\theta})$$

- **Likelihood** $p(\boldsymbol{y} \mid \boldsymbol{\theta})$: function that takes information from the data.
- **Prior distribution** $p(\boldsymbol{\theta})$: represents our previous knowledge about the parameter of interest.
- **Posterior distribution** $p(\boldsymbol{\theta} \mid \boldsymbol{y})$: represent what you know after having seen the data. The basis for inference, a distribution, possibly multivariate if more than one parameter.

Bayesian inference. Principles

- Interpretation of probability is related to our **degree of belief** about the event we are considering.
    - For example, in the case of rain, an 80% probability of rain simply tells us that it is more likely to rain than not to rain.
- Information and uncertainty we have about everything unknown is expressed in **terms of probability distributions**.
- It considers as uncertain elements of the problem not only **the data** but also **the parameters**.
- The idea of **repeated sampling is not used** to interpret the properties of estimators.
- **Observed information updates knowledge about the unknown**.
- **Estimators disappear** and are **inferred in terms of probability distributions**.
- **Frequentist approach relies on sample data (present)**. **Bayesian uses also prior information (past)**.

Example: Scoring penalties Valencia C. F.

- Liga Santander is one of the famous league around the world. In this example, we use data of the last 10 seasons in order to know the chance of success (π) to score a penalty for **Valencia Club de Fútbol**.

Example: Scoring penalties Valencia C. F.

Response variable + Data

- $Y = \text{score/miss the penalty}$
- The model is generated by Y
- **Bernoulli** with parameter π, i.e., $Y \sim Ber(\pi)$
- **Likelihood**

$$p(\boldsymbol{y} \mid \pi) = \ell(\pi) \propto \pi^k (1-\pi)^{N-k}$$

k: times that a player score a penalty (30).

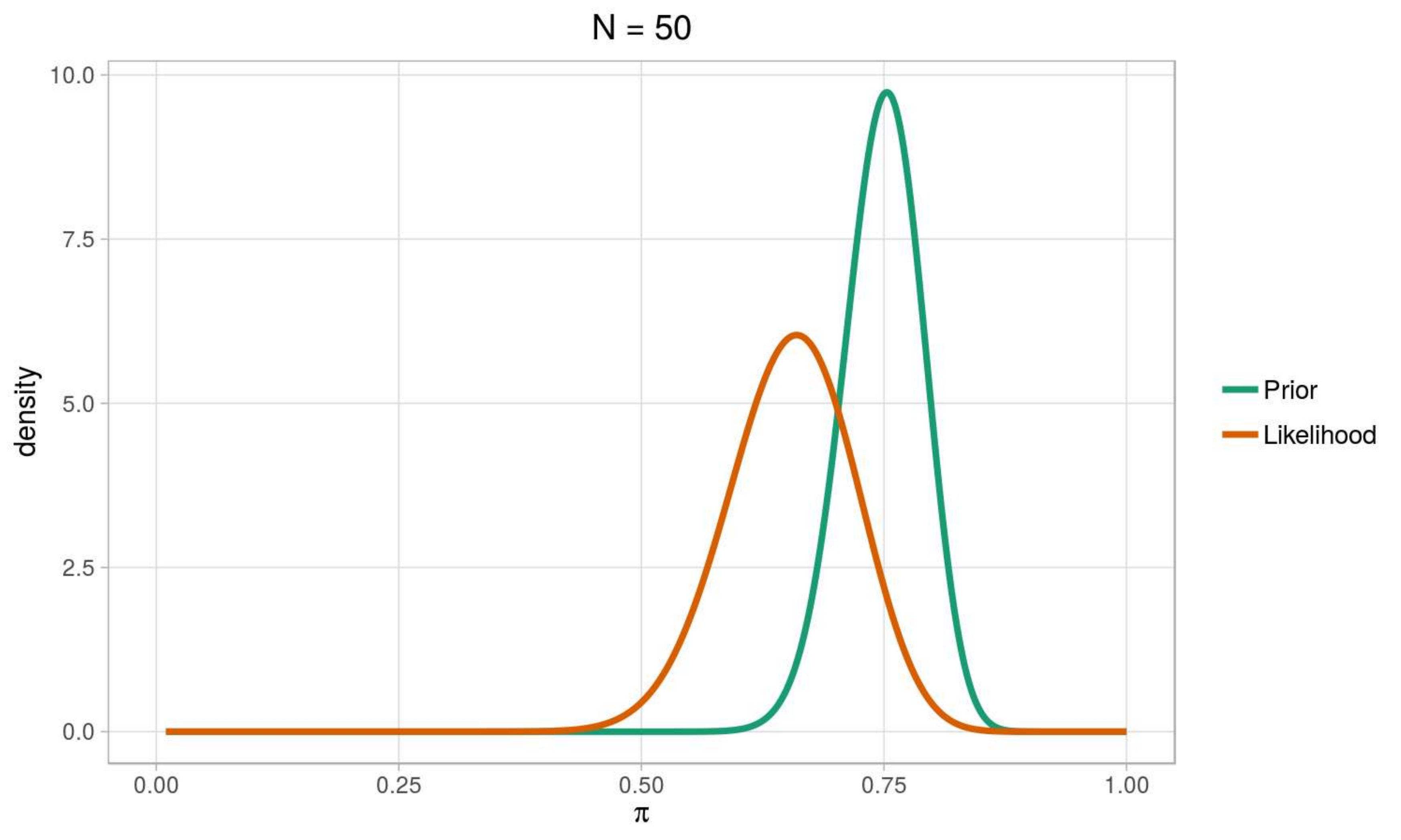
N: total penalties in 10 seasons (50 matches).

Prior knowledge about the parameter π

- **Beta distribution** seems adecuate to model a proportion π.
- After asking some experts, we end up with a 75 percentage chance to score a penalty.
- We express this uncertainty using percentiles $per_{90} = 0.75$ and $per_{50} = 0.75$.
- The corresponding values for a and b are $a = 83.46$ and $b = 28.05$.
- **Prior distribution**

$$p(\pi) \propto \pi^{a-1}(1-\pi)^{b-1}$$

Example. Likelihood *vs* Prior



Graphical Model

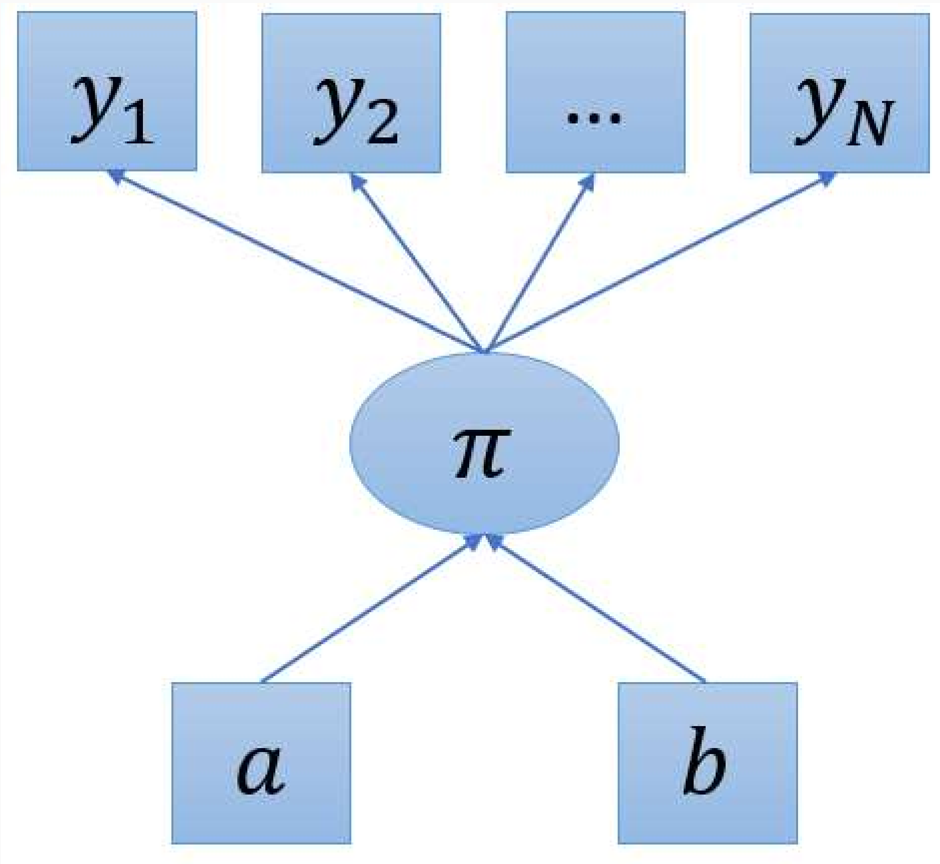
Likelihood

$$p(\boldsymbol{y} \mid \pi) \propto \pi^{k}(1-\pi)^{N-k}$$

Prior distribution

$$p(\pi) \propto \pi^{a-1}(1-\pi)^{b-1}$$

$$\pi \sim \mathrm{Beta}(a, b)$$



Ellipses: variables

Squares: data

Posterior distribution. Bayesian learning process

Estimating the probability to score a penalty

Likelihood

$$p(\boldsymbol{y} \mid \pi) = \pi^k (1-\pi)^{N-k}$$

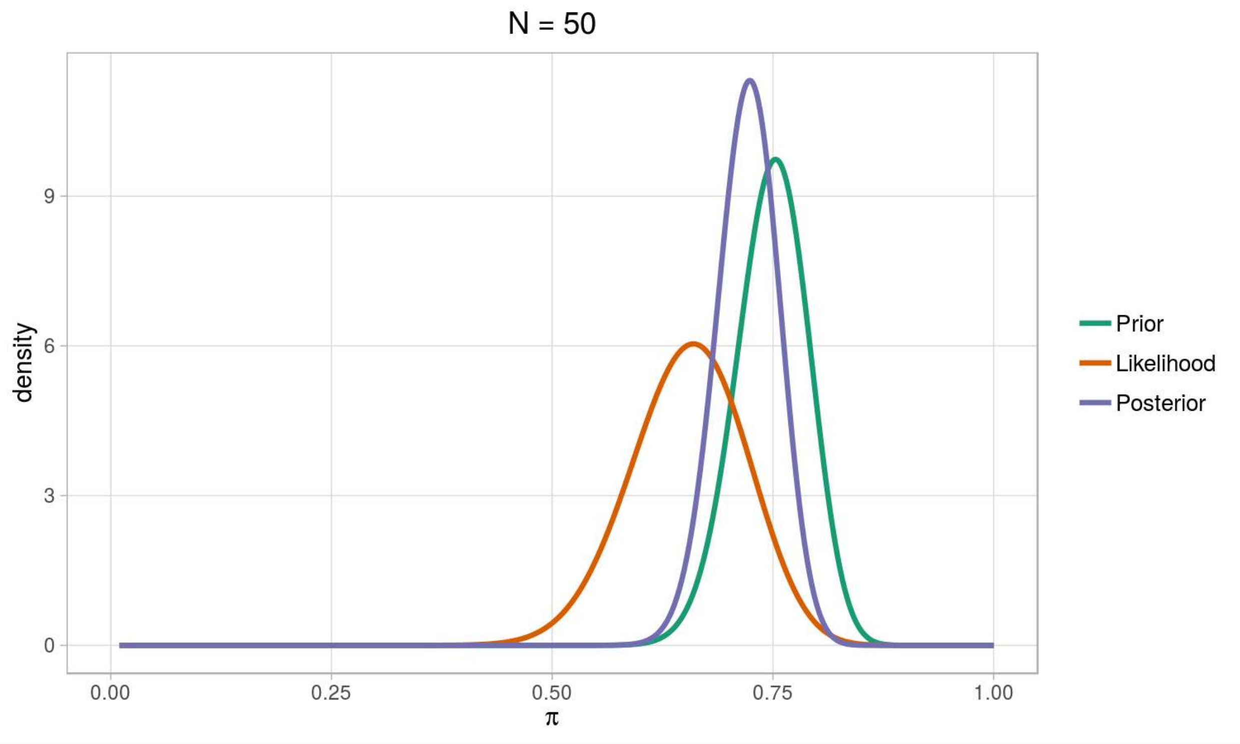
Prior distribution

$$p(\pi) = \pi^{a-1}(1-\pi)^{b-1}$$

Posterior distribution

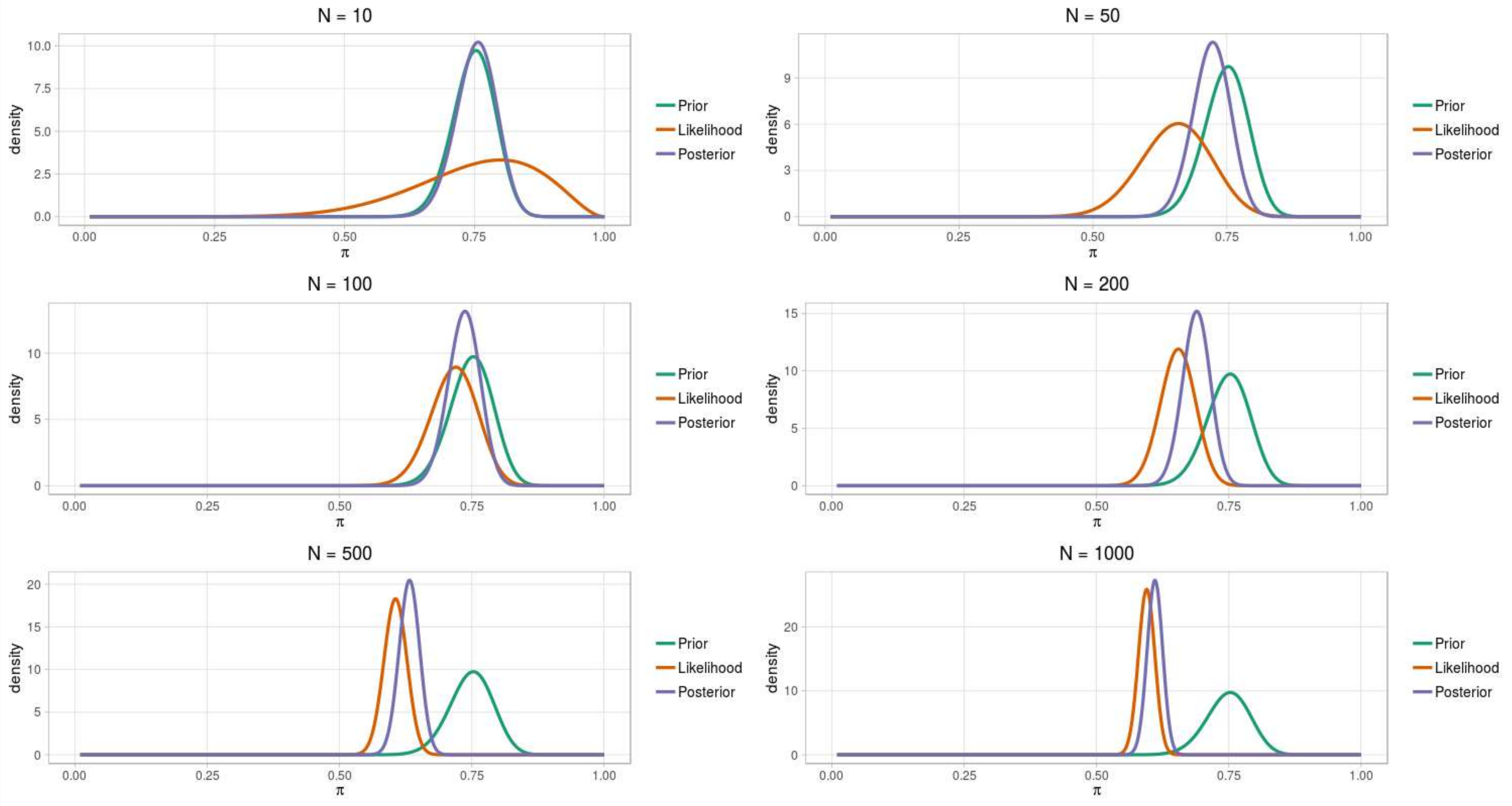
$$\begin{aligned} p(\pi \mid \boldsymbol{y}) &\propto p(\boldsymbol{y} \mid \pi) \cdot p(\pi) \\ &\propto \pi^{k+a-1}(1-\pi)^{N-k+b-1} \end{aligned}$$

$$\pi \mid \boldsymbol{y} \sim \text{Beta}(k+a, N-k+b)$$
$$\pi \mid \boldsymbol{y} \sim \text{Beta}(30+83.46, 20+28.05)$$
$$\pi \mid \boldsymbol{y} \sim \text{Beta}(113.46, 48.05)$$



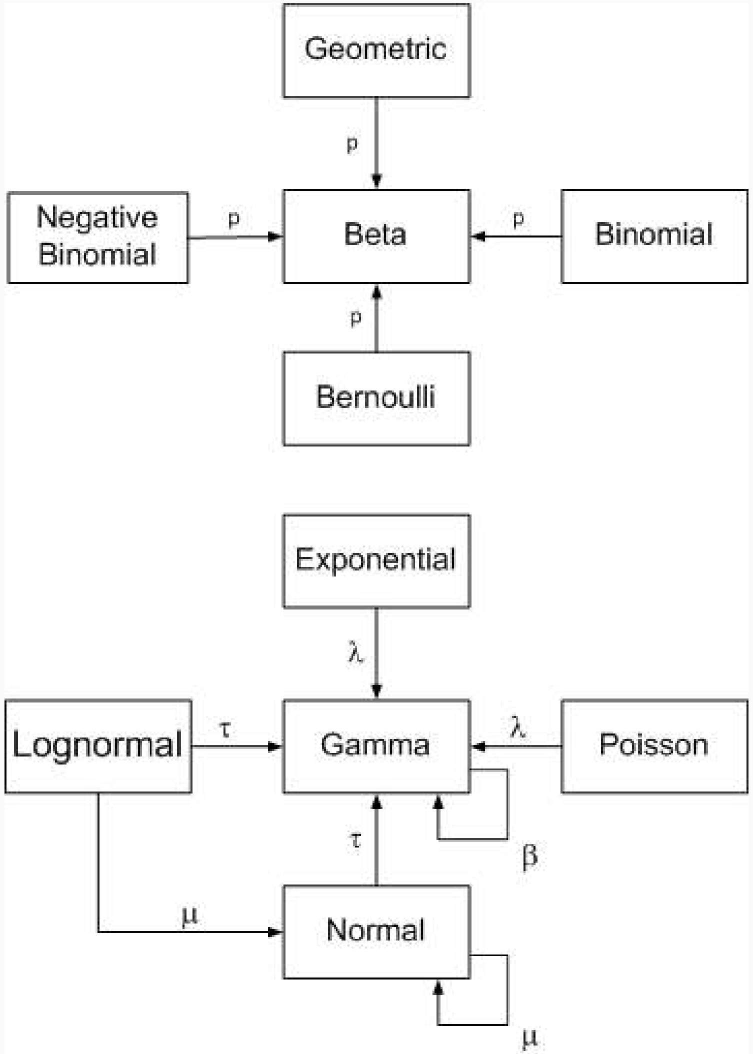
Let's try to understand how a prior works:
https://minaya.shinyapps.io/Beta-Conjugate-Priors/

Data *vs* prior information



Conjugate priors

- **Beta distribution is a conjugate prior** for the Binomial likelihood function.
    - If the posterior distribution $p(\theta \mid y)$ is in the same probability distribution family as the prior probability distribution $p(\theta)$, the prior and posterior are then called conjugate distributions, and the prior is called **a conjugate prior for the likelihood function** $p(y \mid \theta)$.
- Compendium Of Conjugate Priors



Describing results: point estimators, credible

- We obtain a **probability or a density function as a posterior**. So, we can deal with the complete distribution

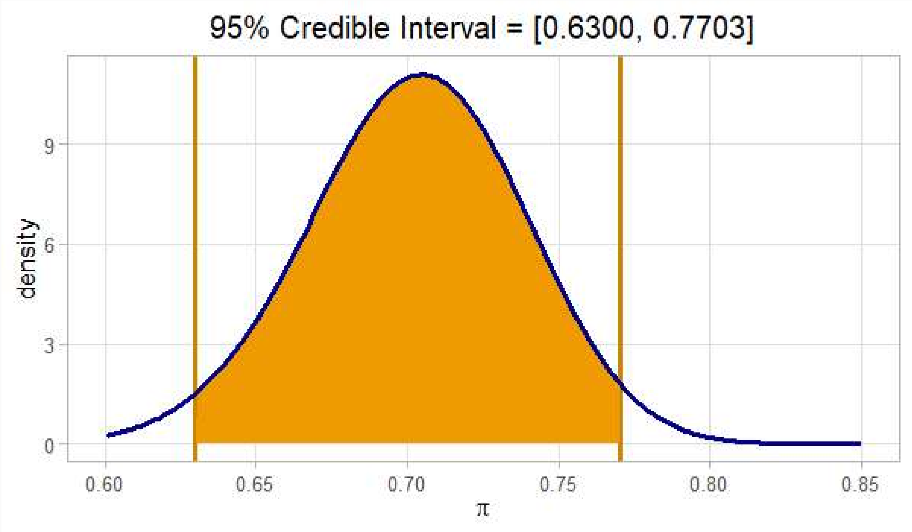
Point estimates

- Mean, median, mode

Credible intervals

- $100(1-\alpha)\%$ credibility interval (CI) for a parameter $\boldsymbol{\theta}$ is defined as the pair of values a and b such as : $p(\theta \leq a \mid \boldsymbol{y}) = \alpha/2$ and $p(\theta \geq b \mid \boldsymbol{y}) = 1 - \alpha/2$

- The $IC_{95\%}(\pi) = (0.63; 0.77)$

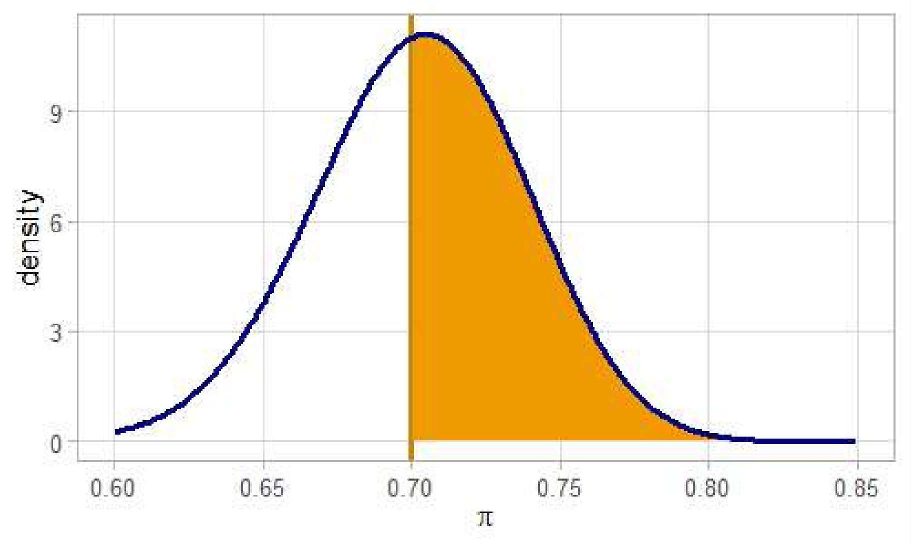


Credible interval *vs* Confidence interval

- **Frequentist approach**: a $100(1-\alpha)\%$ confidence interval is defined such that, if the data collection process is repeated again and again, then in the long run, $100(1-\alpha)$ **of the confidence intervals formed would contain the (fixed) unknown parameter value**.

- **Bayesian approach**: a $100(1-\alpha)\%$ credible interval will explicitly indicate **the posterior probability that θ lies within its boundaries**. So, we talk about credibility intervals.
  - The $IC_{95\%}(\pi) = (0.63; 0.77)$ means that the probability for π to be between 0.63 and 0.77 is 0.95.

- Bayesian inference can also provide any probability statements about parameters.
  - For example, we could compute $p(\pi > 0.7 \mid \boldsymbol{y}) = 0.5368$.



Hypothesis Testing in the Bayesian Framework

- **Hypotheses:**
  - $H_0 : \pi \leq 0.75$
  - $H_1 : \pi > 0.75$
- **Posterior probabilities**
  - $\alpha_0 = P(\pi \leq 0.75 \mid \boldsymbol{y})$
  - $\alpha_1 = P(\pi > 0.75 \mid \boldsymbol{y})$
- **Decision rule**: Choose the hypothesis with higher posterior probability.
  - α_1 quantifies how credible it is, given the data, that $\pi > 0.75$.

- Posterior: $\pi \mid \boldsymbol{y} \sim \mathrm{Beta}(113.46, 48.05)$
- Computed posterior probabilities:
  $P(H_0 \mid \boldsymbol{y}) = P(\pi \leq 0.75 \mid \boldsymbol{y}) = 0.91$
  $P(H_1 \mid \boldsymbol{y}) = P(\pi > 0.75 \mid \boldsymbol{y}) = 0.09$
- **Interpretation**: About 9% of the posterior mass lies above $\pi = 0.75$.
  Therefore, there is **limited posterior evidence** that the true scoring probability exceeds 0.75.

"Bayesian testing directly compares the credibility of each hypothesis via their posterior probabilities."

🔗 *Note:* The **Bayes Factor** is used for **hypothesis testing** and also plays a key role in **model selection** within Bayesian inference.
See the R package BayesVarSel for practical applications.

3. Predictions

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Predictions

Prior predictive distribution

- Using just the **previous information** about the population.
- **Before performing the experiment** one can infer about the most/least probable values to be observed.

$$p(y_{pred}) = \int p(y_{pred} \mid \theta) p(\theta) d\theta$$

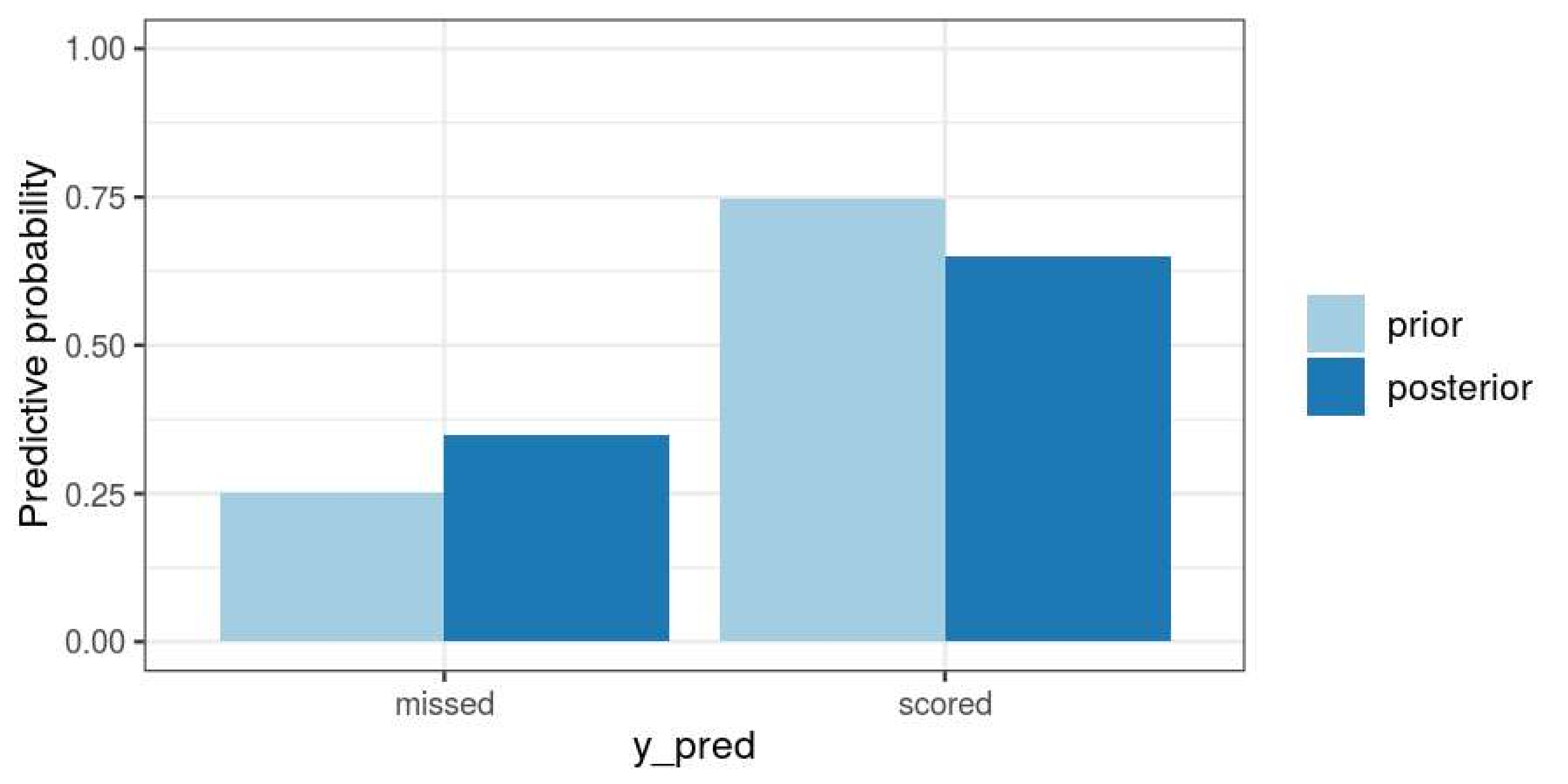
Posterior predictive distribution

- Using the **updated** information after performing the experiment.
- Allows us to infer about the most/least probable values to be observed if we would **repeat the experiment** in the future (in the same conditions).

$$p(y_{pred} \mid y_{obs}) = \int p(y_{pred} \mid \theta) p(\theta \mid y_{obs}) d\theta$$

Prior *vs* Posterior predictive

- Carlos Corberán wants to know if he can trust in their players to score the next penalty. In this figure we see what happened if we take into account just the expert knowledge or the expert knowledge and the data.



What we have learned so far

- **ALL uncertainty** is quantified through probability distributions.
- Bayes theorem (probability calculus) is the tool to **combine several sources of uncertainty**.
- **Prior information is totally separated from information from the data** (use the data twice is forbidden!)
- Prior information **is updated by data information into posteriors**.
- **Posterior distributions combine and quantify** the information/uncertainty about the state of interest and are the pillar blocks of Bayesian inference.

Numerical approaches

- When applying Bayesian Statistics, most of the usual models do not yield analytic expressions for neither the posterior nor the predictive posterior distributions.
- Most of the **complications that appear in the Bayesian methodology** come from the resolution of integrals that appear when applying the learning process:
    - The normalization constant of the posterior distribution,
    - moments and quantiles of the posterior,
    - credible regions, probabilities in the contrasts, etc.

Solutions:

- **Monte Carlo methods: MCMC**.
- **INLA**.

References

Blogs

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- *A Bayesian Approach to Finding Lost Objects* — **Katie Howgate**, *Lancaster University STOR-i Blog*

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